UNIT 5 · RAPID REVISION

Formula and Fact Sheet

Anthropometry and work physiology

Item	Value
Clearance dimensions	Design for the 95th percentile
Reach dimensions	Design for the 5th percentile
Adjustability	Span the 5th–95th percentile
Inside cab height (Liljedahl p. 227)	185 cm
Seat height is set by	Popliteal height
Seat-pan depth is set by	Buttock–popliteal length
1 L of oxygen	≈ 20.9 kJ ≈ 5 kcal
Heart rate → energy (Indian)	E (kJ/min) ≈ 0.159·HR - 8.72
Acceptable workload (8 h)	≈ 35% of VO_2max
Limit of continuous performance	HR ≈ 110 beats/min; ΔHR ≤ ~40
Indian male VO_2max	≈ 2.0 L/min
Sustained force limit	≤ ~30% of maximum voluntary strength
Murrell rest allowance	R = T(b - s)/(b - 1.5)
Thermal comfort (ASHRAE ET*)	24–27 °C

Noise and vibration

Item	Value
Audible range	16–20,000 Hz
Sound pressure level	SPL = 20·log(p/p_0); p_0 = 20 μPa
Two equal sources	+3 dB (90 + 90 = 93 dB)
OSHA limit	90 dB(A) for 8 h; +5 dB halves the time
Impulse noise	≤ 140 dB peak
Most sensitive WBV band (vertical)	4–8 Hz
Tractor bounce frequency	3–5 Hz
Whole-body vibration guide	ISO 2631 (FDP boundaries)
Hand-arm vibration standard	ISO 5349
Seat isolation condition	Frequency ratio r > √2
Tool handle diameter (power grip)	30–40 mm

Safety, chemicals, controls and seating

Item	Value
Red label triangle	Extremely toxic, oral LD_{50} 1–50 mg/kg
Yellow label triangle	Highly toxic, LD_{50} 51–500 mg/kg
Blue label triangle	Moderately toxic, LD_{50} 501–5000 mg/kg (DANGER)
Green label triangle	Slightly toxic, LD_{50} > 5000 mg/kg (CAUTION)
ROPS standards	ASAE S383.1 (ROPS); S310.3 (overhead protection)
Thresher safety standards	IS 9020 (safety); IS 9129 (feeding chute — now within IS 9020)
Seat-pan angle / backrest angle	5° / ≈ 95° (min 90°)
Leg–thigh angle for pedals	110–120°
Seat adjustment	≥ 150 mm fore-aft; ≥ 100 mm vertical
Seat suspension	f_n < 1.5 Hz; travel ≈ 76 mm
Foot clutch (ASAE S335.3)	Left foot, forward/downward to disengage
Priority instrument zone	30° cone — 380 mm wide at 700 mm

Anthropometry, physiological cost, fatigue (Topics 1–3)

- Clearance → 95th percentile; reach → 5th; adjustability 5th–95th; cab inside height 185 cm (Liljedahl p. 227); SAE J833 anthropometric practice; popliteal height → seat height.
- 1 L O_2 ≈ 20.9 kJ ≈ 5 kcal; E (kJ/min) ≈ 0.159·HR - 8.72; AWL = 35% VO_2max; continuous-work HR limit ≈ 110; ΔHR ≤ 40; Indian male VO_2max ≈ 2 L/min.
- Christensen: light 2.5–5, moderate 5–7.5, heavy 7.5–10 kcal/min. Static work fatigues faster (occlusion). Murrell R = T(b - s)/(b - 1.5).
- ODR (10-point), BPDS (Corlett–Bishop map); ET* comfort 24–27 °C; cab pressurization at 50 Pa cuts dust ~6× (146 → 24.7 mg/m³).

Tools, noise and vibration (Topics 4, 6)

- Ear range 16–20,000 Hz; SPL = 20 log(p/p_0), p_0 = 20 μPa; dB(A) = ear-weighted; equal sources +3 dB.
- OSHA: 90 dB(A) for 8 h, 5-dB halving rule, impulse ≤ 140 dB; cabs cut operator noise ~94 → ~80 dB(A) (Nebraska data, Liljedahl p. 207).
- WBV: most sensitive vertical 4–8 Hz; tractor bounce 3–5 Hz; ISO 2631 FDP; spine deformations in 71.3% of tractor drivers (Rossegger, p. 212); ride limits usable power (Matthews, 2/3).
- Seat: transmissibility 0.4–0.65; seat f_n ≈ 0.4–0.5 × tyre f_n; isolation only beyond r = √2. HAV → white finger (ISO 5349).
- Hand tools: handle Ø 30–40 mm; long handles beat bending; ≤ 30% MVC sustained; CIAE improved tools = more output, lower HR/ODR.

Safety, chemicals and PPE (Topics 5, 7, 8)

- Safety by education, regulation and DESIGN — design that removes the hazard is most positive (Liljedahl p. 234).

- USA 1982: overturns 49% of tractor fatalities; run-over 20% (over half fell off first); rate 17.6 → 8.1 per 100,000 tractors (1969–82) with ROPS/belts (p. 231).
- ROPS: skeleton frame / frame + roof / enclosed cab; ASAE S383.1 & S310.3; tests = static or dynamic → crush → field upset → temperature–material; useless without the seat belt.
- Indian machine hazards: thresher cylinder (IS 9129 feeding chute; IS 9020 safety), PTO entanglement (master shield + tunnel guard), nip points, hydraulic injection injuries.
- Label triangles: RED ≤ 50, YELLOW 51–500, BLUE 501–5000, GREEN > 5000 mg/kg oral LD_{50}; lower LD_{50} = more toxic; dermal is the main spraying route; OP-poisoning antidote = atropine.
- Hierarchy: eliminate → engineering → administrative → PPE last; nitrile, not cotton, gloves; observe the waiting interval; triple-rinse containers.

Controls, vision and seating (Topics 9–10)

- Seat: pan angle 5°; backrest–seat ≈ 95° (min 90°) with lumbar support; leg–thigh 110–120°; foot–leg ≥ 90°; adjust ≥ 150 mm fore-aft and 100 mm vertical; armrests vertically adjustable (Liljedahl pp. 226–227).
- Seat standards: SAE J899 (dimensions), J1163 (location), J898 (comfort/reach zones), J185 (steps/handrails), J154a (space envelope), J209 (instrument faces).
- Suspension: f_n < 1.5 Hz and 0.4–0.5 × tyre f_n; travel ≈ 76 mm; transmissibility 0.4–0.5; softer = better isolation but more movement relative to controls; largest tyres ride best.
- Controls: frequently used → RIGHT-hand zone, left hand stays on steering; PTO/parking brake/diff lock → left/foot outside zone; ASAE S335.3 uniform location & motion — clutch = LEFT foot, forward/downward to disengage; symbols ASAE S304.5/SAE J389b; colour coding EP443.
- Vision: near-ground front & rear + far all round; priority instruments inside the 30° cone, 380 mm wide at 700 mm; engine gauges left of centre, transmission right.

Further reading

- Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 9 (Human Factors in Tractor Design).
- Gite & Yadav / CIAE Bhopal — anthropometric and physiological data of Indian agricultural workers (AICRP on Ergonomics and Safety).
- ISO 2631 (whole-body vibration), ISO 5349 (hand-arm vibration); OSHA noise criteria.
- ASAE S383.1, S310.3, S335.3, S304.5, EP443; SAE J833, J899, J1163, J898, J185, J154a, J209.
- BIS IS 9020 (thresher safety); Insecticides Act 1968 & Rules 1971; CIB&RC list of banned/restricted pesticides.
